MUKUND SHANKAR

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EDUCATION

M.S. in Computer Science, *University of Maryland* Expected December 2026

B.S. in Computer Science, **B.S. in Mathematics**, *University of Maryland* May 2025
Minor: Robotics & Autonomous Systems, **Honors:** 6× Dean's List

EXPERIENCE

Machine Learning Engineer Intern *JuumeAI* April 2025 - September 2025

- Built and owned a production PostgreSQL/pgvector RAG system over **10K+** documents, with automated ingestion, LLM-based semantic chunking, and APIs consumed by frontend engineers; deployed on Google Cloud Run and improved BERTScore by **17%** over direct state-of-the-art LLM baselines.
- Reduced document indexing time by **5**× through multithreading and autoscaling, and built CI/CD, health checks, route validation, and monitoring to keep ingestion and retrieval services reliable in production.
- Developed a stage-wise evaluation framework using BERTScore, RAGAS, and coverage metrics to diagnose retrieval failures, validate grounding, and measure performance throughout the RAG pipeline.

Research Assistant *Dr. Mohit Iyyer, UMD CS* April 2026 - Present

- Investigating long-form narrative generation in LLMs using SFT and RLHF, with emphasis on coherence, stylistic consistency, and human-like storytelling across long-form outputs.

Undergraduate Researcher *Dr. Abhinav Shrivastava, UMD CS* July 2023 - November 2025

- Designed and trained a transformer-based text-to-SMPL model, achieving **90% recall accuracy** on KitML and HumanML3D benchmarks.
- Extended the model to RGB videos via keypoint detection and SAM tracking to build a generalizable end-to-end motion/text retrieval pipeline evaluated on **39,000 videos**.

PROJECTS

FullTimeBrief *Co-Founder & Engineer* — Website June 2026 - Present

- Built and operate a live football platform end-to-end across frontend, backend, databases, deployment, monitoring, and an agentic editorial pipeline that discovers, ranks, verifies, and prepares 5–12 stories/day.
- Engineered a fault-tolerant near-live match pipeline that reconciles multiple data sources every 60s, supporting 10+ concurrent matches and edge cases including overturned goals, delays, extra time, and rescheduling.

Diagnosing Tool-Call Decision Making in Small Language Models *Python, PyTorch, Transformers, LoRA Tool-Calling LLMs, Model Probing, Post-Training* Code — Report

- Investigated decisions among tool use, clarification, and abstention on the When2Call benchmark, comparing prompting, SFT, DPO, and Constitutional AI-style supervision on Gemma-3-4B-it and Llama-3.2-3B-Instruct.
- Improved tool-call accuracy over zero-shot by **32.8%** (Llama, SFT) and **24.3%** (Gemma, CAI-DPO).
- Trained linear probes over hidden states to diagnose representation-behavior gaps, showing that correct tool-call decisions were often internally recoverable despite incorrect final outputs.

Confidence Probes for Language Models *Python, PyTorch, Transformers, scikit-learn QA Systems, LLM Correctness* Code — Report

- Built lightweight MLP probes from hidden states, entropy, and layer trajectories to predict LLM answer correctness, achieving **90% accuracy** and **0.97 AUC-ROC**.
- Trained probes on **424K examples** across TriviaQA, HotpotQA, SQuAD v2, GSM8K, and MMLU, evaluating cross-dataset and cross-model transfer on Llama 3.1 8B and Qwen 2.5 models from 1.5B–14B; identified threshold brittleness and task mismatch under distribution shift.